

The strong humidity sensitivity observed in our trials likely masked more nuanced trends that were otherwise expected, making it difficult to isolate other contributing mechanisms. To resolve this, paired exposures using dry, well-characterized synthetic air and ambient humid air could be used to compare conductivity, pressure responses, and gas-switch transients, revealing which effects persist without moisture. Additionally, redesigning the electrode geometry—by narrowing spacing and increasing gold surface area—could test whether electrode-mediated H_2 splitting or surface-area-driven processes are responsible for the doping effects reported in the literature. Combined with time-series measurements and surface-sensitive techniques such as XPS, these approaches could help disentangle humidity-related, electrode-driven, and intrinsic aging mechanisms, offering pathways to stabilize and enhance CuSCN conductivity for solar applications.